

Only 2/12 held-out systems are dual-class; 10/12 are single-class (hatched n/a).

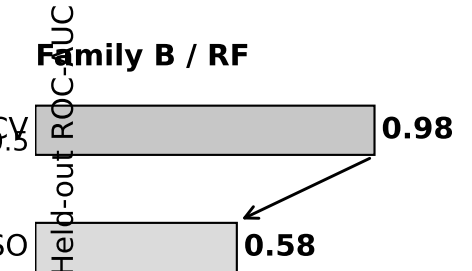
Family B
(CV → LOSO collapse)

Key reading

Dual-class folds: 2/12
hotel_booking, login_system
Only these rows support
stable ROC-AUC under LOSO.

Single-class: 10/12 systems
(undefined ranking; hatched n/a)

Family B (graph tallies), RF:
CV 0.982 → LOSO μ 0.583
 $\Delta \approx 0.40$



Cell value = mean held-out ROC-AUC
across LR, DT, and RF.

n/a (single-class)

Held-out requirement system (system_id)

dual-class
folds

hotel_booking	1.00	0.33	0.89	0.78
login_system	0.67	0.67	1.00	1.00
access_control	n/a	n/a	n/a	n/a
atm	n/a	n/a	n/a	n/a
bike_rental	n/a	n/a	n/a	n/a
elevator	n/a	n/a	n/a	n/a
package_locker	n/a	n/a	n/a	n/a
parking_gate	n/a	n/a	n/a	n/a
smart_thermostat	n/a	n/a	n/a	n/a
train_ticket_booking	n/a	n/a	n/a	n/a
vending_machine	n/a	n/a	n/a	n/a
warehouse_inventory	n/a	n/a	n/a	n/a

10 single-class (n/a)

A: gates

B: graph tallies

C: ref.-diff.

D: combined

Predictor family (A--D)

Family B spread: 0.33–0.67